Contact & Address Book Manager

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Objective and method

Improve the assigned address book using Pylint, Bandit and Flake8. The original demo and source were preserved, the tools were run before editing, and the final source was analyzed again with the same rules. Security fixes, quality fixes and formatting fixes were separated into three commits.

Measured results

Check	Before	After
Pylint	5.47/10; 29 messages	10.00/10; no messages
Bandit	4 findings: 1 High, 1 Medium, 2 Low	0 findings
Flake8	4 violations	0 violations
Behavior checks	Original demo saved	12 tests passed

Analysis environment: Python 3.12.14; Pylint 4.1.3; Bandit 1.9.4; Flake8 7.4.1. Scope: address_book.py only. No diagnostic suppressions or custom exclusions were used.

Verification

The regression suite checks phone validation, replacement/defaults, name and phone search, city updates/grouping, unexpected errors, sorted output, CSV quoting and Unicode, JSON round trips, invalid/missing backups, deterministic IDs, import behavior and the full demo. The demo output matches the original except for the changed hash ID.

Compatibility and evidence

contacts.bak is now JSON rather than pickle; old backups must be regenerated. IDs remain short display labels and are not guaranteed unique. Function names now use snake_case, and the demo runs only when the script is executed directly. Restore still returns data without replacing the global address book.

Evidence files: reports/before/, reports/after/, reports/test_results.txt and original/. The empty final Flake8 report means no violations; exit code 0 is recorded in run_metadata.json. Full commit IDs and a recoverable Git bundle accompany the submission.

Issues identified and corrected

All line numbers below refer to the preserved original source. Bandit severity is tool-reported; quality and style findings are not assigned invented severity levels.

Issue / type	Original lines	Description and fix
B324 | High / Security	19	MD5 used for contact ID. Use SHA-256; retain six hexadecimal display characters. Not an authentication token.
B301 | Medium / Security	65-67	pickle.load can execute code from a malicious backup. Use json.load/json.dump with UTF-8 text files. Regenerate old pickle backups.
B403 | Low / Security	3	Import of pickle introduces an unsafe serialization facility. Remove pickle and import json.
B105 | Low / Security	6	Hardcoded password literal. Remove unused SYNC_PASSWORD; no synchronization feature needs it.
W0718 / Quality	34	Catching Exception hides unexpected programming errors. Catch KeyError for a missing contact; allow unrelated failures to surface.
W1514 / Portability	54	CSV output relies on the platform default encoding. Specify encoding="utf-8"; keep newline="" for CSV.
C0200 / Quality	49-50	range(len(names)) requires repeated indexing. Use enumerate(names, start=1) and access the contact by name.
C0103 / Naming	10-65	camelCase function and argument names. Use snake_case consistently in definitions and all calls.
C0114 / C0116 / Documentation	1; 10-70	Module and function docstrings are missing. Describe module purpose and every function contract.
E225 / Formatting	23	Missing spaces around assignment. Change found=[] to found = [].
E303 / Formatting	47	Three blank lines before a top-level function. Keep two blank lines between top-level functions.
E231 / Formatting	84	Missing space after a comma. Insert the missing space before sorted(restored).
E501 / Formatting	85	A line has 86 characters, beyond Flake8 limit 79. Wrap the final print call using parentheses.

The additional main guard prevents demonstration output and file creation when importing the module for tests. It is an extra improvement, not a finding attributed to the three tools.

Reflection and reproducibility

Reflection draft for student review

The supplied handout refers to reflection questions but does not include them. The following four prompts address the learning outcomes; they are not quoted instructor questions or a claim of personal experience.

1. How do the three tools complement one another?

Pylint identified maintainability problems such as broad exception handling, missing documentation and naming. Bandit detected security-sensitive patterns, including pickle deserialization and MD5. Flake8 identified spacing, blank-line and line-length violations. Using all three covered different aspects of the same source file.

2. Which findings mattered most, and why?

Unsafe pickle loading was the most direct risk because a modified backup could cause code execution when loaded. Bandit marked MD5 as High and pickle loading as Medium, so both were addressed first. Here the hash is only a display ID; replacing MD5 does not turn a six-character ID into a secure or collision-free identifier.

3. What tradeoffs and challenges arose while fixing the program?

Replacing pickle required changing both backup and restore together. JSON preserves the string-and-dictionary contact data, but existing pickle backups cannot be loaded by the new code. Narrowing the exception to KeyError preserves the missing-contact message while exposing unexpected TypeError failures. Renaming functions required updating every demonstration call.

4. How were the fixes verified, and what are the limits of static analysis?

The final source passed all three tools and 12 regression checks. The demo output matches the baseline except for the deliberately changed SHA-256 display ID. Tests cover normal and invalid input, search, city changes, CSV quoting and Unicode, and backup round trips. Clean tool output does not prove correctness or security; JSON content is not schema-validated, CSV fields are not neutralized for spreadsheet formulas, and more realistic applications would need stronger input and file protections.

Reproduce the results

Create a Python virtual environment, install requirements-dev.txt, and run the following from the submission folder. The complete Windows commands are in SUBMISSION_README.md.

```
python -m pylint --persistent=n address_book.py
```

```
python -m bandit -r address_book.py -f txt
```

```
python -m flake8 address_book.py
```

```
python -m unittest -v
```

Source and authorship note

Assigned source: https://github.com/SETAPESU26/L5_27_address_book; baseline 12f84cd2821f7a74712fa8755f095758391c4fac. Assignment and identity mapping came from the supplied lab handout and repository-assignment spreadsheet. Prepared with AI assistance; review and personalize the reflections before submission.